Ritabrata Chakraborty

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Education

University of Pennsylvania — **Pennsylvania, USA**
M.S.E. in Robotics, ***GPA: –/4*** — *Aug '26 – May '28*

Birla Institute of Technology and Science, Pilani (BITS Pilani) — **Rajasthan, India**
B.E. in Mechanical Engineering — Minor in Data Science, ***CGPA: 8.57/10*** — *Oct '22 – May '26*

Publications & Peer Reviews

Journal Publications

- **R. Chakraborty** and K. Kishore, ***"Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation."*** Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.
- **R. Chakraborty** and K. Kishore, ***"Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments."*** Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, **R. Chakraborty**, and G. Sartoretti, ***"CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction."*** Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**.
- **R. Chakraborty**, T. Mian, and P. Kundu, ***"An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery."*** Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**.
- K. Kishore and **R. Chakraborty**, ***"Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring."*** Publication awaiting intellectual property clearance, 2025.

Poster Presentations

- **R. Chakraborty**, ***"High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun."*** Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

Peer Reviews

- ***"GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments."*** Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- ***"Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD)."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"A Research Paper on IoT-Based Robot for Agriculture."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

Research Experience

Research Intern — **BITS Pilani**
Supervisor: Dr. Avinash Gautam, Associate Professor — *May '25 – Aug '25*

- Literature study of vision-language grounded multi-robot coordination and VLMaps-based navigation.

Research Intern — **CSIR-CEERI**
Supervisor: Dr. Kaushal Kishore, Principal Scientist — *Jan '24 – May '26*

- Monocular vision-based UAV navigation for orchard monitoring using YOLOv11.
- Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation.

Research Intern | **MARMoT Lab, NUS**
Supervisor: Dr. Guillaume Sartoretti, Assistant Professor | *Mar '25 – Dec '25*

- Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan).
- Vision-attention-driven autonomous navigation with semantic understanding, extending CogniPlan and ViNT.

Research Assistant | **BITS Pilani**
Supervisor: Dr. Mani Shankar Dasgupta, Associate Professor | *Apr '24 – Oct '25*

- Physics-based launcher mechanical design and FEM analysis.
- Computer vision pipeline for 3D trajectory reconstruction and shot classification using PnLCalib.
- Finite element analysis of running track surfaces using hyperelastic material models.

Research Intern | **KU Leuven**
Supervisor: Dr. Pradeep Kundu, Assistant Professor | *Sep '24 – May '26*

- Synthetic data generation for bearing fault diagnosis under data scarcity.
- Physics-informed generative modeling for bearing fault diagnosis under data scarcity, validated on LiteFormer.

Research Assistant | **BITS Pilani**
Supervisor: Dr. Rajneesh Choubisa | *Aug '24 – Dec '24*

- Primal world beliefs study surveying engineering undergraduates on core worldview dimensions.

Research Intern | **BARC**
Supervisor: Dr. Rishi Verma, Head (ESDS), PPEMD | *Jun '24 – Jul '24*

- Image-based visual servoing for automated radar control and UAV tracking using YOLOv8 and DeepSORT.
- Impact testing and analysis of metal-composite shields using electromagnetic railguns.

Industry Experience

ML Intern | **Uber**
Supervisors: Mr. Siddarth Malreddy, Tech Lead Manager & Mr. Ishan Nigam, Senior ML Engineer | *Jul '25 – Dec '25*

- Object annotation automation and anomaly detection in uLabel.

Subject Matter Expert | **Knowledge Cultivator**
Self-supervised | *Nov '23 – Feb '24*

- Technical problem-solving and academic content authoring using MATLAB, CAD, and Python.

Projects

Vision-Guided Robotic Arm for Simulated Oral Cancer Screening | *Aug '25 – Dec '25*
Autonomous Drone Navigation | MathWorks Global Student Drone Challenge 2025 | *Mar '25 – Apr '25*
Air Quality Index Forecasting Using Time Series Models | *Nov '24 – Nov '24*
3D Lunar Surface Modeling from DEM Data | ISRO Immersion Challenge 2024 | *Jul '24 – Jul '24*
Cricket Team Optimization | AMEX Campus Challenge 2024 (Product Track) | *Jun '24 – Jul '24*
ExoMy Rover Navigation and UR3 Arm Motion Planning | ERC 2023 Remote | *Apr '23 – Sep '23*

Teaching Experience

Teaching Assistant
CIS 4190/5190: Applied Machine Learning, University of Pennsylvania | *Aug '26 – Dec '26*
CS F320: Foundations of Data Science, BITS Pilani | *Jan '26 – May '26*
ME F218: Advanced Mechanics of Solids, BITS Pilani | *Jan '25 – May '25*
ME F216: Materials Science and Engineering, BITS Pilani | *Sep '24 – Dec '24*

Honours & Scholarships

Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students) | *Oct '22 – May '26*

Awards & Achievements

Finalist — Meta PyTorch OpenEnv Hackathon (India) | *Apr '26*
3rd Place — MathWorks Global Drone Student Challenge 2025 | *Mar '25*
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) | *Jul '24*
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 | *Jul '24*
5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals) | *Sep '23*

Leadership Experience

Deputy Director of Academic Programming
Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania *Aug '26 – Present*

Vice-President
South Asian Association (Rangoli), University of Pennsylvania *Aug '26 – Present*

President & Secretary
Mechanical Engineering Association (MEA), BITS Pilani *Jun '24 – May '26*

President & Tech Fest Coordinator
Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani *Oct '24 – Jul '25*

Project Manager
Tinkerer's Lab (TL), BITS Pilani *May '24 – Jul '25*

Treasurer
Bengali Cultural Association (Moruchhaya), BITS Pilani *May '24 – Jul '25*

Volunteering Experience

Volunteer
Junoon, National Service Scheme (NSS), BITS Pilani *Sep '23 – Sep '23*

Professional Activities

Graduate Student Member — IEEE *2026*

Competitive Examinations

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3) *2025*
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28) *2025*
JEE Advanced: All India Rank 14372 *2022*

Technical Skills & Coursework

Relevant Coursework: Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins

Programming Languages: Python, C++, C, C#, Java, Bash (Shell)

Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS

AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel

Cloud & MLOps: AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions

Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR

CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360

Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing

Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace